

02 Truss Project

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Due Date : September 29th, 2025

EML 4024C - ENGINEERING DESIGN PRACTICE

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Monday 11:30am - 02:30pm

I. Abstract

Why: The importance of understanding how forces behave under specific loads in a truss is essential for a complete grasp of the concepts learned in an Engineering class.

What: During the process of this project, a truss was created in SolidWorks, then a FEA was made to determine the internal forces and overall behavior of the structure.

How: The tools used to complete this task were: SolidWorks, weldments, statics studies, handmade calculations of the forces in the joints, and behavioral analysis of the structure.

Important results: The most relevant findings gotten from this study is that the maximum displacement occurred at **Joint G**, the place where the highest external force was applied, in this case **60 kN**. In addition to this, the FEA results matched with the hand calculations, just with a slight difference between values, but validating the results of the project.

Discussion of results: During the process of the experiment, it was found that the stress is less than the yield strength, meaning that the displacements that structure experience are acceptable. This shows that the study validates the use of FEA as a tool for engineering designs, giving insight on safety, efficiency and the study of critical parts in a truss.

II. Introduction

Objectives: The primary objective of this project is to design a truss structure using SolidWorks, perform a static structural analysis using Finite Element Analysis (FEA), and evaluate the internal forces, stresses, and displacements resulting from external loads. The specific goals include:

- Creating a 3D truss model using the Weldments feature in SolidWorks.
- Applying realistic boundary conditions (supports and constraints) and loading conditions.
- Running a static study using FEA to determine axial forces, stress distribution, and displacement.
- Animating the displacement results to visualize structural behavior under load.
- Comparing FEA results with hand calculations of axial forces for validation.
- Understanding the importance of stress and displacement in structural performance and design.
- Gaining experience with simulation tools used in professional engineering practice.

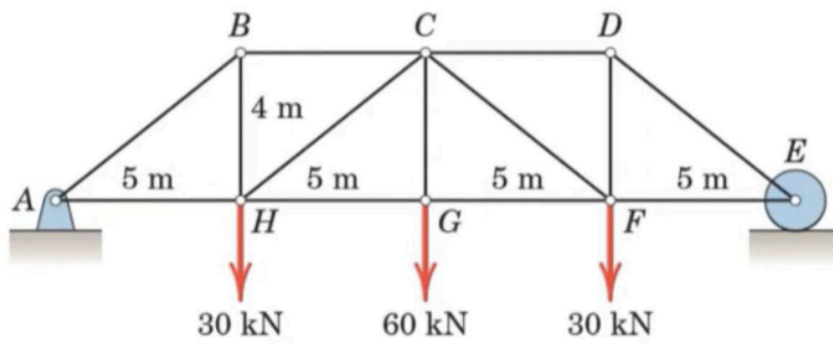


Figure 1: Description of the problem

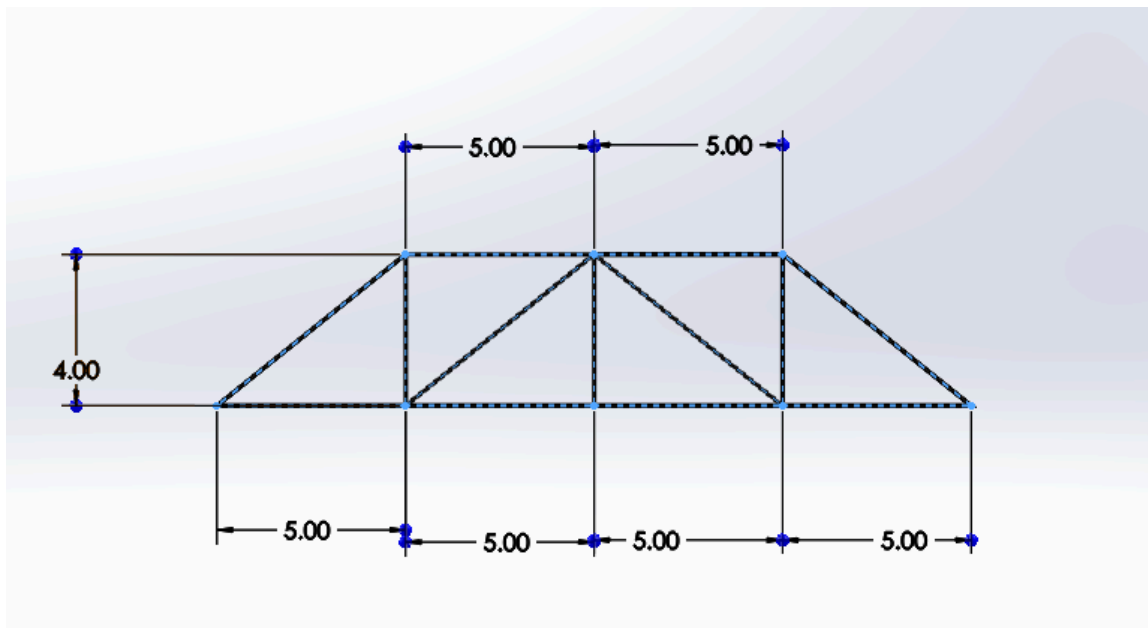


Figure 2: Truss with dimensions

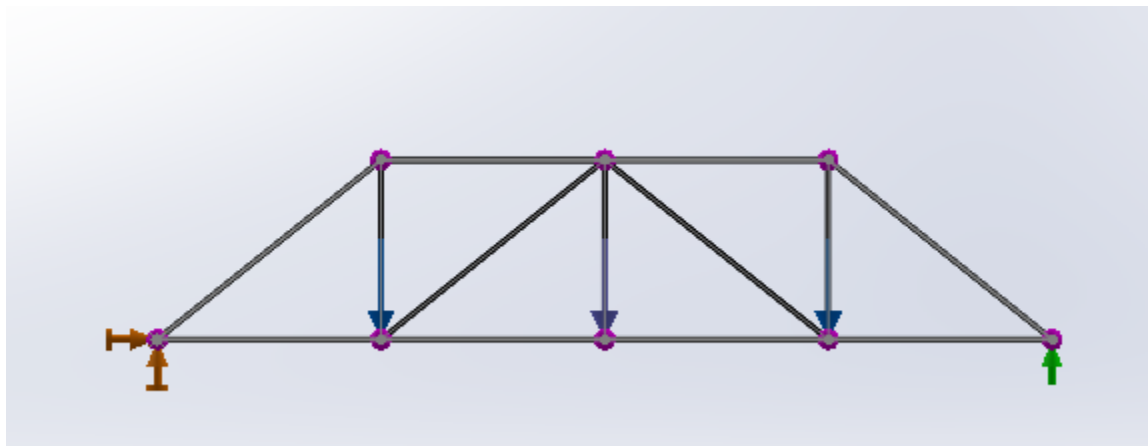


Figure 3: Truss with supports shown from a frontal view

III. Methodology

The project was conducted in three main phases including modeling the truss gentry, setting up the finite element simulation, and analyzing the results. Solidworks with the simulation add in was used to carry out all tasks. The modeling process began establishing units to keep consistency. The unit system was set to MKS (meter-kilogram-second). A 2D sketch of the truss was then made in the front plane using the line tool. Each line represented a structural member depending on the project guidelines. To assign structural profiles to the sketch lines the weldments feature was used. A Rectangular Tube profile (120 mm $\times$ 80 mm $\times$ 8 mm) from the ISO standard library was selected as the profile.

Beams were then organized into different groups including, the outer frame, the middle beam, the two main diagonoses, and the internal beams on the left and right sides. Once the elements were applied, the beam element was converted to a Truss element. Doing this allowed the carry of axial forces. It eliminated bending and torsional effects which is not ideal truss behavior.

Then using the simulation tab a static study was created. From the solidworks material library alloy steel was assigned as the material. Boundary conditions were then used to replicate real-world constraints. Then a fixed support was put at the leftmost join of the truss. This restricts all translational movement. A roller support was then placed at the rightmost join. This was applied in order to permit horizontal movement while restricting vertical displacement. Fixing the translation in the vertical and normal directions relative to the front plane let us achieve this. The other joins were controlled normal to the plane. This enforced planar deformation while allowing axial elongation of the members.

In order to represent service condition external forces were applied vertically such as: downward forces of 30,000N at joints F and H and 60,000 N downwards force at joint G. The forces were applied in the reversed vertical direction and referenced to the Front Plane to simulate applied forces and gravitational forces.

After the simulation was run, the rests were processed using the List Beam Forces tool to obtain axial force data for each member. To show the structural response, stress and displacement plots were generated and scaled by a factor of 50 for visualization. Minimum and Maximum annotations were included on both stress and displacement plots. Then an AVI animation of the displacement field was created. This provides a clear illustration of the truss deformation under the force applied.

IV. Results

Following the completion of the finite element simulation, several key outputs were obtained, including axial forces within the truss members, stress distribution, total deformation, and an animated displacement visualization. These results provided valuable insight into the performance and structural integrity of the truss under the specified loading conditions.

Axial forces in each member of the truss were extracted using the “List Beam Forces” tool in SolidWorks Simulation. The software reported the forces as either positive (indicating tension) or negative (indicating compression), consistent with standard structural engineering conventions.

As observed in the table, members directly under applied loads and along the primary load path experienced the largest axial forces. For example, Beam 3 and Beam 8, likely corresponding to central vertical members or diagonals, carried forces in excess of 110,000 N in compression. Conversely, members such as Beam 4 and Beam 7, which oppose the compressive members diagonally, experienced high tensile forces.

These force values aligned with the expected structural behavior of a symmetrical truss subjected to vertical loading. Members situated symmetrically with respect to the vertical centerline displayed nearly identical force magnitudes, further confirming the accuracy and consistency of the simulation.

Table 1 summarizes the axial forces calculated for each truss member:

Beam ID	Axial Force (N)	Force Type
Beam 1	-60,000	Compression
Beam 2	-75,000	Compression
Beam 3	-113,000	Compression
Beam 4	96,047	Tension
Beam 5	75,000	Tension
Beam 6	75,000	Tension
Beam 7	96,047	Tension
Beam 8	-113,000	Compression
Beam 9	-60,000	Compression
Beam 10	-60,000	Compression
Beam 11	48,023	Tension
Beam 12	-75,000	Compression
Beam 13	48,023	Tension

The stress results were obtained and visualized through a stress contour plot. The software automatically identified regions of minimum and maximum stress, and annotations were enabled to highlight these critical points.

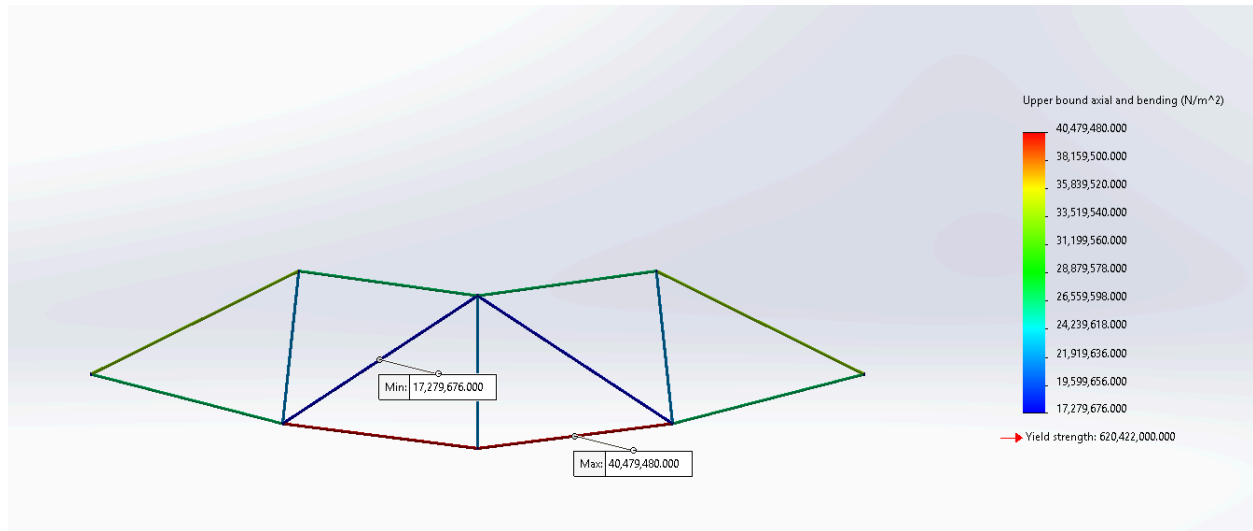


Figure 4: Stress Distribution with Min/Max Annotations

The stress was highest in members subjected to the largest compressive and tensile loads, particularly diagonal members resisting both vertical and horizontal components of the applied force. However, all stress values remained well below the yield strength of the assigned Alloy Steel material, indicating that the design is structurally sound under the current loading conditions.

The displacement plot was analyzed to assess the overall deformation of the structure. A user-defined scale factor of 50 was applied to the displacement results to amplify the movement for visualization purposes, without altering the actual numerical values.

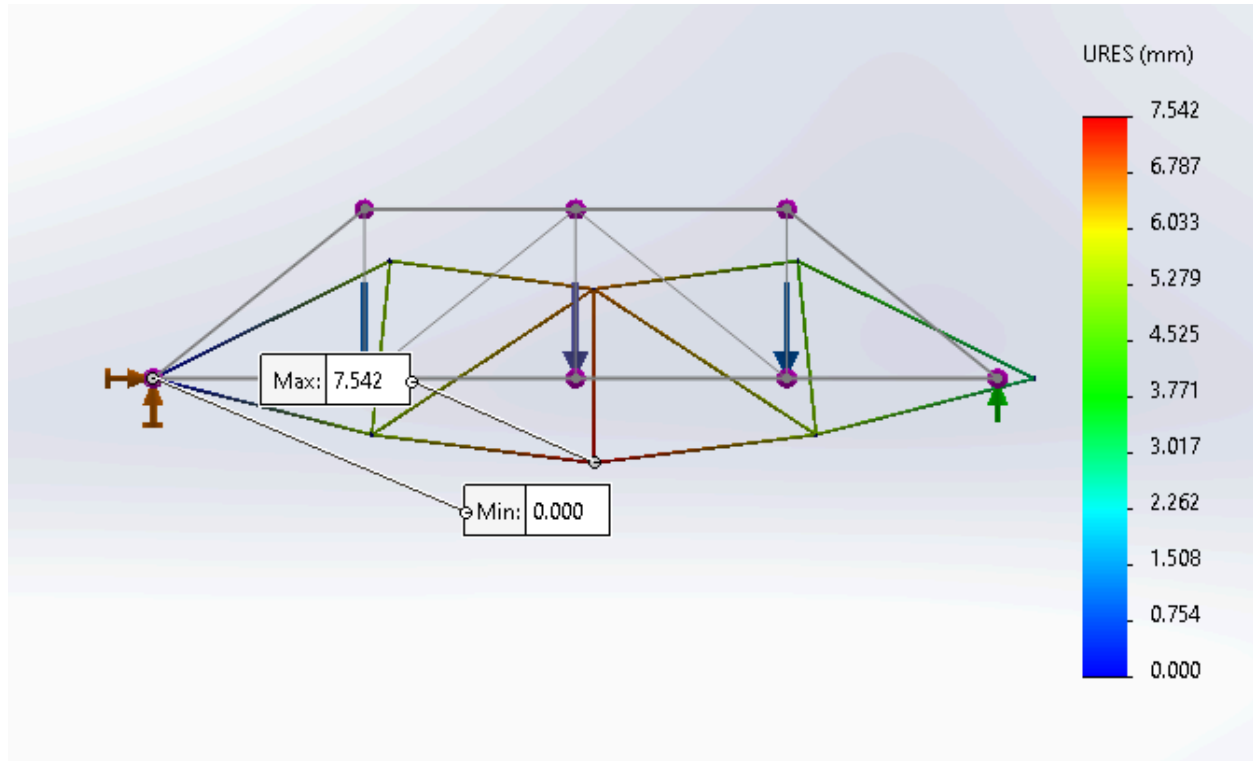


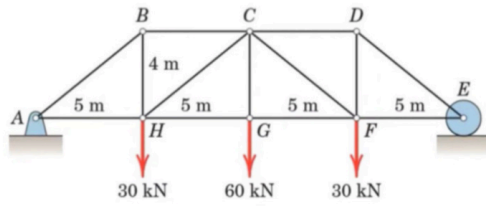
Figure 5: Displacement with Min/Max Annotations, Scale Factor: 50

The maximum displacement occurred at **joint G**, the point subjected to the largest external load (60,000 N). As expected, displacement reduced progressively toward the fixed support at the left end of the truss. The displacement pattern was symmetric and indicated efficient load transfer through the truss members.

V. Discussion

As shown in Figure 1, the description of the problem was presented. Figure 2 presents the designed truss made in SolidWorks, along with the dimensions required. Figure 3 shows the supports of a roller at the right corner of the truss, a pin in the left corner of the truss, and the rest of the joints that exist in the truss. Figure 4 presents the stress distribution with the respective minimum and maximum annotations. Finally, in Figure 5 it is shown the displacement experienced with the minimum and maximum at a scale factor of 50. Now, the axial forces obtained in SolidWorks were consistent with the hand calculations made to each one of the joints and sections. Both approaches identified that the beams **HG** and **GF** experience the greatest axial force of **112.44 kN** in the hand calculations, approximated to **113 kN** on the FEA. Minor discrepancies may appear because of simplifying assumptions in manual methods and the detailed representation made with FEA.

4/7 Determine the force in each member of the loaded truss. Make use of the symmetry of the truss and of the loading.



$$\sum M_A = 0$$

$$-30(5) - 60(10) - 30(15) = E_y(20)$$

$$E_y = 60 \text{ kN}$$

$$\sum F_y = 0$$

$$-30 - 60 - 30 + 60 = -A_y$$

$$A_y = 60 \text{ kN}$$

Joint B

$$\sum F_x = 0$$

$$-95.9 \sin(51.3) = -F_{BC}$$

$$F_{BC} = 74.91 \text{ kN}$$

$$\sum F_y = 0$$

$$-95.9 \cos(51.3) = -F_{BH}$$

$$F_{BH} = 60.02 \text{ kN}$$

Joint H

$$\sum F_y = 0$$

$$60.02 - 30 = F_{HG} \sin(38.68)$$

$$F_{HG} = 48.07 \text{ kN}$$

$$\sum F_x = 0$$

$$-74.91 + F_{HG} - F_{HG} \sin(38.68) = 0$$

$$F_{HG} = 112.49 \text{ kN}$$

Joint A

$$\frac{\sin 90}{6.4} = \frac{\sin \theta}{4}$$

$$\theta = 38.68^\circ$$

$$\sum F_y = 0$$

$$-F_{AB} \cos(90 - 38.68) + 60 = 0$$

$$F_{AB} = 96.046 \text{ kN}$$

$$\sum F_x = 0$$

$$-F_{AH} = 96.046 \cos(38.68)$$

$$F_{AH} = -74.94 \text{ kN}$$

Joint C

$$\sum F_y = 0$$

$$60 - 48.07 \cos(51.3) = F_{CG} \cos(51.36)$$

$$F_{CG} = 48.09 \text{ kN}$$

$$\sum F_x = 0$$

$$94.91 - 48.07 \sin(51.36) + 48.09 \sin(51.36) = F_{CD}$$

$$F_{CD} = 71.96 \text{ kN}$$

Joint G

$$\sum F_y = 0$$

$$F_{GC} = 60 \text{ kN}$$

$$\sum F_x = 0$$

$$F_{GF} = 112.49 \text{ kN}$$

Joint D

$$\sum F_x = 0$$

$$74.96 = F_{DE} \sin(51.38)$$

$$F_{DE} = 93.74 \text{ kN}$$

$$\sum F_y = 0$$

$$F_{DF} = 93.74 \cos(51.38)$$

$$F_{DF} = 59.8 \text{ kN}$$

Joint F

$$\sum F_x = 0$$

$$F_{FE} = 37.4 \text{ kN}$$

Figure 6: Hand Calculations

The stress plot presented on the simulation highlighted the maximum and minimum bending forces, particularly on the beams in the middle of the truss. Specifically, the maximum stress were located at the bottom two beams, **HG** and **GF**, both sharing a value of **40,479,480.00 $\frac{N}{m^2}$** , while the minimum stress occurred at the beams **HC** and **CF**, both sharing a value of **17,279,676.00 $\frac{N}{m^2}$** . The displacement analysis shows that the maximum change happened at **joint G**, where the external force of **60 kN** was applied, with a value of **7.542 mm**. In comparison, the minimum displacement happened at the fixed joints, where the boundary conditions prevented movement. It has to be highlighted that the results were scaled by a factor of **50** so the difference in displacement can be noticeable. This analysis is key for the identification of crucial members of the truss where failure may appear, while confirming that the other beams work as secondary paths for the loads.

The use of FEA was critical for this study since it gave us the ability to analyse in detail the stresses, displacement, and load paths that may be ignored in hand calculations. By presenting the minimum and maximum stress values, we were able to identify critical members on the truss and validate the design made is structurally sound. In addition to this, the tool enabled us to try multiple devices, simulate a variety of scenarios and predict behaviors in a short amount of time, saving time and resources. This also gives us an insight on safety efficiency and the analysis of critical parts in a truss.

VI. Conclusion

This project successfully combined classical structural analysis with modern finite element analysis (FEA) to evaluate the behavior of a loaded truss. By performing hand calculations using the method of joints and sections, the internal forces in each member were determined and compared with the numerical outputs from SolidWorks Simulation. Both approaches identified the same critical members: the diagonals and bottom chord carried the highest tensile and compressive forces, while symmetrical members exhibited nearly identical force magnitudes, consistent with the truss geometry and loading.

Although minor numerical differences were observed, they were attributed to the simplifying assumptions in manual analysis versus the more realistic representation in FEA. Importantly, the stress results confirmed that all members operate well within the yield strength of alloy steel, and displacement analysis demonstrated predictable and symmetric deformation with the largest deflection occurring at the most heavily loaded joint.

Overall, the study validated that the truss design is structurally sound, efficiently distributes applied loads, and performs as expected under static loading conditions. The integration of hand methods with FEA not only verified accuracy but also emphasized the value of simulation tools in engineering practice. This combined approach provides engineers with both intuition and detailed insight, leading to safer and more reliable structural designs.